

TORIN NELSON

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Research Interests: scientific ML, computational science, ML systems, HPC, and AI for scientific/engineering applications.

SKILLS

Languages: Python, C/C++

Machine Learning & Scientific Data: PyTorch, scikit-learn, NumPy, pandas, Machine Learning, Statistics, Quantitative Analysis, NLP, Scientific Computing

Tools & Engineering: Git, Linux, REST APIs, FFmpeg, Test-Driven Development, Technical Documentation

EDUCATION

University of Colorado, Boulder

Expected Mar 2027

MSCS in Computer Science (AI Specialization) | 4.0 GPA

University of California, Irvine

Jun 2024

B.S. in Computer Science (AI Specialization) | 3.7 GPA

PROJECTS

Computational Tone-Mapping Utility - Python

Jan 2024 - Present

- Engineered a high-performance Python/FFmpeg utility for GPU-accelerated video processing and mathematical tone mapping.
- Designed a thread-safe, asynchronous rendering pipeline with real-time tracking, machine fingerprinting, and Azure code signing.
- Used Test-Driven Development with unittest.mock to maintain over 90% test coverage across core processing paths.
- Developed a GPU-accelerated processing architecture, increasing encoding throughput by 400% via hardware-specific optimizations for Intel, AMD, and NVIDIA GPUs.

Quantitative Analysis & Statistical Signal Processing - Python

Nov 2025 - Jan 2026

- Built an automated real-time statistical analysis system in Python for ingesting, scoring, and filtering high-frequency probability signals.
- Implemented a historical variance filtering model to analyze probability distributions and filter out non-systematic noise signals.
- Developed a multithreaded polling architecture with variance estimation, slippage estimation, and consensus weighting for signal confidence scoring.

Attention UNet Remote Sensing - Python

Jan 2024 - Mar 2024

- Developed an Attention UNet model using satellite remote sensing data to map underserved regions in Sub-Saharan Africa.
- Achieved 75% segmentation accuracy and used explainable AI techniques (XAI) to interpret spatial feature representations.
- Received the Future Prospects and Recognition Award for adapting a medical imaging model to satellite data.

EXPERIENCE

Sankaku Analytics

Feb 2026 - May 2026

Founding Engineer and CTO - Remote

- Developed the MVP for a high-performance data analytics platform, focusing on scalable backend architecture and efficient data processing.
- Implemented and optimized K-Means clustering algorithms for automated data segmentation and pattern discovery.
- Architected the core system from the ground up, making critical decisions on tech stack and data integrity to support rapid prototyping and iteration.
- Presented findings to CEO, Head of Retail, and Data Analyst using their proprietary data.

Deputy Project Manager and CDH Systems Lead

Apr 2024 - Sep 2024

NASA L'SPACE - Remote

- Co-led a 12-person team through a NASA mission concept which simulated NASA phases A and B to analyze rock-ice samples on PSR region(s) on the Moon, delivering a 240-page PDR and revising the mission scope to fit a simulated 22% budget cut.
- Defined CDH subsystem requirements, including variable data transmission rates (30s Science / 1s Telemetry) for critical mission phases.
- Conducted trade studies for avionics and mechanical components (IMU, Motors), selecting hardware based on thermal operating ranges (-193°C to 111°C) and power budgets.